The University of Nottingham
Malaysia Campus

SCHOOL OF COMPUTER SCIENCE

A LEVEL 1 MODULE, AUTUMN SEMESTER 2016-2017

SYSTEMS AND ARCHITECTURE

Time allowed ONE Hour

Candidates must ***NOT*** *start writing their answers until told to do so*

Answer TWO questions out of THREE

Only silent, self-contained calculators with a single line display are permitted in this examination.

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Dictionaries are not allowed with one exception. Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination. Subject specific translation dictionaries are not permitted.

No electronic devices capable of storing and retrieving text, including electronic dictionaries, may be used.

DO NOT turn examination paper over until instructed to do so

1. **Networks (25 Marks)**

(a) A cable TV company is considering to replace all coaxial cables with optical fibre cables. List any two advantages and two disadvantages of this plan.
[4 Marks]

(b) What is a token, and how are tokens used to control network access? [3 Marks]

(c) An Ethernet LAN with 3 workstations and a Token Ring LAN with 4 workstations are to be internetworked. Twisted pair cables will be used to connect between workstations and other relevant devices within the LAN.

Draw a network diagram indicating the type of devices that are needed to enable machines in both LANs to exchange information as if they are in a single logical LAN.
[3 Marks]

(d) A computer engineer designs a switch that was made using bridges. In order to save money he/she uses repeaters instead of bridges. Would you buy one of these switches? Why, or why not? [3 Marks]

(e) With the help of diagrams, briefly explain the operations of a circuit switching network. List any one advantage and one disadvantage of this approach.
[5 Marks]

(f) Distinguish between adaptive and non adaptive routing algorithms. Why is adaptive routing superior to non adaptive routing?
[3 Marks]

(g) Briefly describe the hidden and exposed station problems in wireless LAN.
[4 Marks]

2. Digital Logic, Data representation, Registers and Memory (25 Marks)

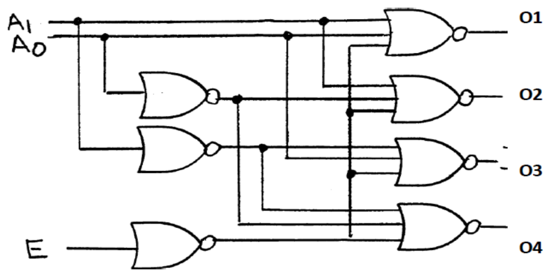
(a) Simplify the following Boolean function in product-of-sums form by using four-variable Karnaugh map. Draw the logic diagram with AND and OR gates.

$F(A, B, C, D) = \Sigma\,(0, 2, 8, 9, 10, 11, 14, 15)$ [5 Marks]

(b) Simplify the algebraic expression using Boolean algebra.

$F = X'\,Y'\,Z + X'\,Y\,Z' + X\,Y'\,Z' + X\,Y\,Z$ [3 Marks]

(c) Write the output generated by the following logic circuit.



[4 Marks]

(d) Convert the following decimal numbers ***a*** and ***b*** into binary and represent the value of ***a + b*** in single (32 bits) and double precision (64 bits) IEEE 754 floating point format as shown in the following diagrams.

***a* = 32** ***b* = 0. 375**

Single precision format

1	8 bits	23 bits
Sign bit	Exponent in excess 127	Mantissa fraction

Double precision format

1	11 bits	52 bits
Sign bit	Exponent in excess 1023	Mantissa fraction

[5 Marks]

(e) Briefly describe the term "*cache coherency*". [4 Marks]

(f) List any four differences between a ***CISC*** processor and ***RISC*** Processor.

[4 Marks]

3. Interrupts, I/O and Instruction Pipelining (25 Marks)

(a) List and briefly define the three techniques for performing I/O. [6 Marks]

(b) Describe the various steps needed for carrying out an interrupt I/O process originated by one or more I/O devices.

[5 Marks]

(c) A processor executes an instruction in the following five stages. The time required by each stage in picoseconds (1,000 ps = 1 ns) is given for each stage.

IF	instruction fetch	300 ps
OF	Operand fetch	250 ps
E	Execute	350 ps
M	Memory access	700 ps
OS	Operand store (write back)	200 ps

(i) What is the time to execute an instruction if the processor is not pipelined?

[2 Marks]

(ii) What is the time taken to fully execute an instruction, assuming that this structure is pipelined in five stages and that there is an additional 20 ps per stage due to the pipeline latches?

[2 Marks]

(d) (i) What is a data hazard? Name and describe a hardware technique that overcomes most data hazards.

[3 Marks]

(ii) Consider the following instruction sequence in a five-stages pipeline IF, OF, E, M, OS:

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1. ADD r0,r1,r2
2. ADD r3,r0,r5
3. STR r6,[r7]
4. LDR r8,[r7]
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Identify all the potential data hazards in the above codes. [3 Marks]

(e) What, if any, is the difference between a multicore processor and a coprocessor?

[4 Marks]